ASSIGNMENT 5

Date: 16/10/2026

1. Aim

To study the importance of spacers in FET fabrication and understand their formation, patterning, and role in advanced semiconductor device structures. To analyze spacer deposition and patterning in the Spacer and 14nm FinFET process flows using Semulator3D.

2. Tool Used

Semulator3D:

3. Theory (Spacer)

Introduction to Spacer:

A spacer is a thin insulating layer formed on the sidewalls of the gate structure in MOSFET and FinFET devices. It is generally made of dielectric materials such as silicon nitride (Si_3N_4) or silicon dioxide (SiO_2).

Spacers are formed by depositing a dielectric layer over the gate structure, followed by an etching process that removes material from horizontal surfaces while retaining it along the vertical sidewalls.

Importance and Applications of Spacer:

Spacers play an important role in semiconductor device fabrication. Their major applications include:

- **Source/Drain Formation:** Help control the position and separation of source/drain regions from the gate.
- **Implantation Mask:** Act as a mask during ion implantation to control dopant placement.
- **Short Channel Effect Control:** Help in controlling source/drain extension dimensions in scaled transistors.
- **Capacitance Reduction:** Reduce unwanted parasitic capacitance between the gate and source/drain regions.
- **Device Performance:** Help maintain proper device dimensions and improve transistor performance.

Spacers are widely used in modern MOSFET and FinFET fabrication, particularly in advanced technology nodes where precise control over transistor dimensions is essential.

Assignment Question 1

Process Editor Steps:

The following table shows the process steps used in the Spacer.zam project.

Step	Action	Step Name	Material(s)	Depth	Mask	Polarity
1	Wafer Setup	Wafer Setup	/Silicon/Si_Xtal	100	–	–
2	Sequence	Fin Formation	–	–	–	–
2.1	Conformal Deposit	Silicon Nitride Hard Mask	/Nitrides/Si3N4	20	–	–
2.2	Planarizing Deposit	Resist for SiN Hard Mask	/Organics/Resist	20	–	–
2.3	Expose Material	Expose Resist	/Organics/Resist	20	Fin_Pattern	Light
2.4	Enhanced Basic Etch	SiN Hard Mask Etch	/Nitrides/Si3N4	20	–	–
2.5	Remove Material	Remove Resist	/Organics/Resist	–	–	–
2.6	Taper Etch	Etch for Fin Formation	/Silicon/	60	–	–
2.7	Remove Material	Remove SiN Hard Mask	/Nitrides/Si3N4	–	–	–
3	Sequence	STI	–	–	–	–
3.1	Planarizing Deposit	Planar Deposit	/Oxides/SiO2	0	–	–
3.2	Enhanced Basic Etch	Basic Etch	/Oxides/SiO2	30	–	–
4	Sequence	Gate Metal Patterning	–	–	–	–
4.1	Conformal Deposit	Gate High-K	/Oxides/HfOx	2	–	–
4.2	Conformal Deposit	Gate Metal Deposition	/Metals/TiN	5	–	–
4.3	Planarizing Deposit	Gate Metal Deposition	/Metals/W	20	–	–
4.4	Planarizing Deposit	Resist for Gate Patterning	/Organics/Resist	100	–	–
4.5	Expose Material	Expose Resist	/Organics/Resist	0	Gate	Light
4.6	Enhanced Basic Etch	Metal Etch	/Metals/TiN	60	–	–
4.7	Remove Material	Remove Resist	/Organics/Resist	–	–	–
5	Sequence	Spacer Pattern	–	–	–	–
5.1	Conformal Deposit	Spacer Deposition	/Nitrides/Si3N4	spacer_thk	–	–
5.2	Enhanced Basic Etch	Spacer Etch	/Nitrides/Si3N4	spacer_thk*1.1	–	–

Spacer Formation

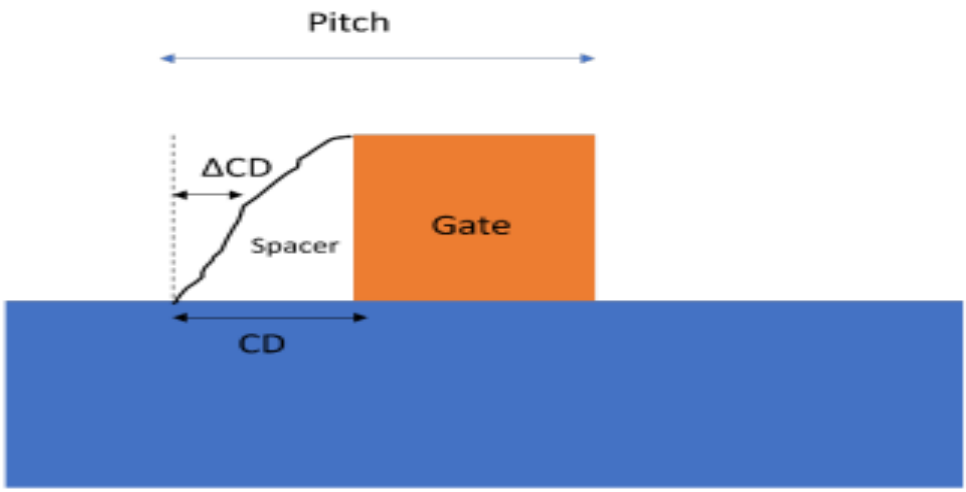


Figure 1: Reference Image

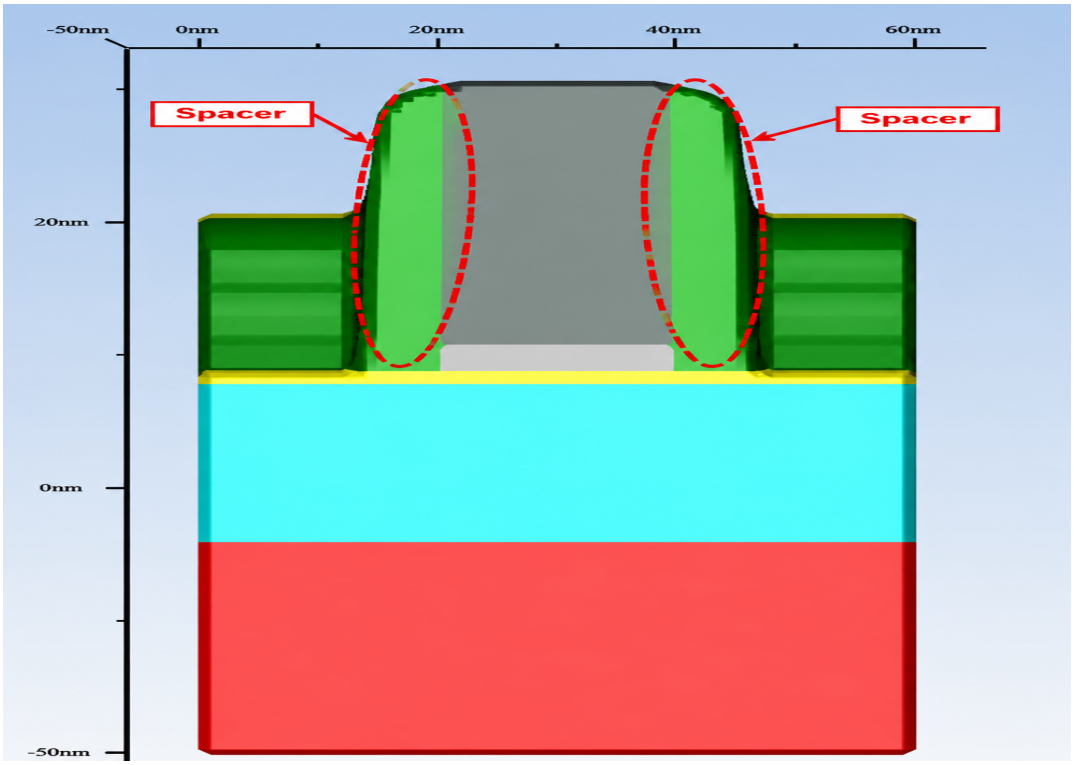


Figure 2: Simulated Result

Assignment Question 1

(a) Deposition Technique Used for Spacer:

The deposition technique used for spacer formation is **Conformal Deposition**. In this process, a thin layer of silicon nitride (Si_3N_4) is deposited uniformly over the gate structure, covering both horizontal and vertical surfaces.

Importance of Conformality:

Conformality is important because it ensures uniform thickness of the deposited layer over the entire structure, including the vertical sidewalls. This helps in obtaining properly shaped and uniform spacers on both sides of the gate.

Common Conformal Deposition Techniques:

- Chemical Vapour Deposition (CVD)
- Low Pressure Chemical Vapour Deposition (LPCVD)
- Atomic Layer Deposition (ALD)
- Plasma Enhanced Chemical Vapour Deposition (PECVD)

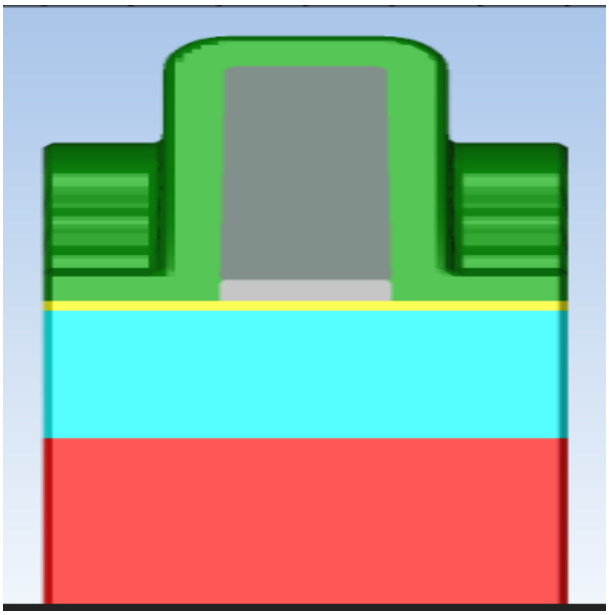


Figure 3: Spacer Deposition

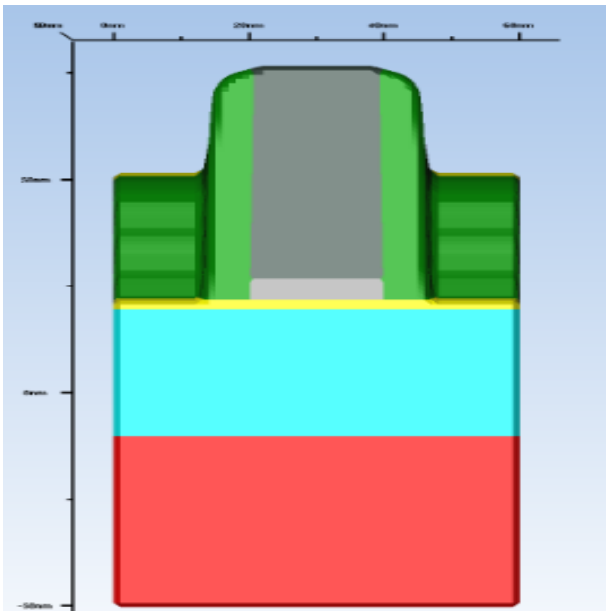


Figure 4: Spacer Etching

(b) Why is Spacer Patterned Without Lithography?

Spacer patterning is performed without lithography because it uses an anisotropic etching process. After conformal deposition, the deposited material is etched in the vertical direction. This removes the material from horizontal surfaces while retaining it on the vertical sidewalls of the gate.

Therefore, spacers are formed automatically on both sides of the gate without requiring an additional lithography mask. This process is known as **self-aligned spacer formation**.

(c) Calculation of CD Difference at 50% Height:

The target CD is 7 nm, whereas the simulated CD width obtained from the process simulation is 7.01171875 nm. The CD at 50% of the total spacer thickness is 6.649137126980349 nm.

Total spacer thickness:

$$T_{\text{spacer}} = 53.83402490615845 \text{ nm}$$

Height at 50% of the total spacer thickness:

$$H_{50\%} = \frac{50}{100} \times T_{\text{spacer}}$$

$$H_{50\%} = 26.9170124531 \text{ nm}$$

CD difference is calculated as:

$$\Delta CD = CD_{\text{simulated}} - CD_{50\%}$$

$$\Delta CD = 7.01171875 - 6.649137126980349$$

$$\boxed{\Delta CD = 0.362582 \text{ nm}}$$

(d) CD Profile Along the Spacer Height:

The CD profile represents the variation in critical dimension along the spacer height. The actual height at each position is calculated using the total spacer thickness.

$$H = \frac{\text{Height (\%)}}{100} \times T_{\text{spacer}}$$

The CD difference at each height is calculated as:

$$\Delta CD = CD_{\text{simulated}} - CD_{\text{at given height}}$$

Sr. No.	Height (nm)	CD (nm)	ΔCD (nm)
1	2.691701	6.972015	0.039704
2	5.383402	6.902577	0.109142
3	8.075104	6.745773	0.265946
4	10.766805	6.738281	0.273438
5	13.458506	6.732538	0.279180
6	16.150207	6.718109	0.293610
7	18.841909	6.703679	0.308040
8	21.533610	6.689249	0.322470
9	24.225311	6.673857	0.337861
10	26.917012	6.649137	0.362582
11	29.608714	6.625379	0.386340
12	32.300415	6.588619	0.423099
13	34.992116	6.546088	0.465631
14	37.683817	6.451959	0.559760
15	40.375519	6.376195	0.635524
16	43.067220	6.275944	0.735774
17	45.758921	6.132113	0.879606
18	48.450622	5.918727	1.092992
19	51.142324	4.983500	2.028218

Table 1: CD Variation Along the Spacer Height

CD Profile Graph:

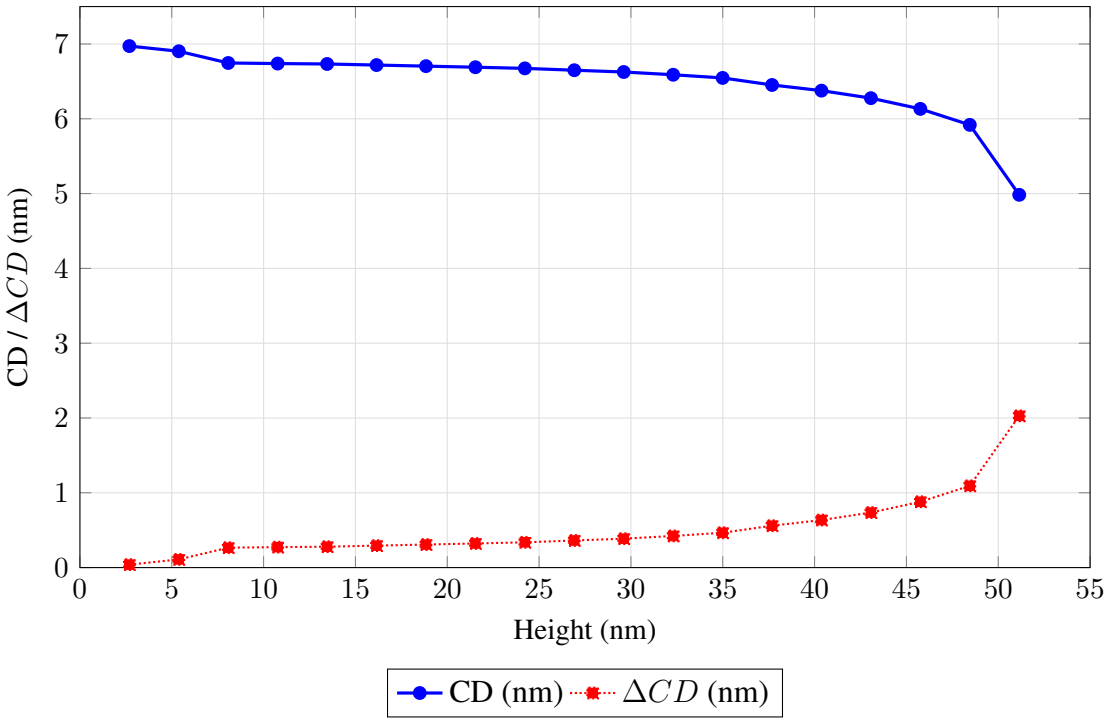


Figure 5: CD and ΔCD Variation Along the Spacer Height

4. Question 2: Process Analysis of 14 nm FinFET

Process Editor:

The following table represents the process flow of the 14 nm FinFET structure, including the major fabrication sequences and their individual process steps.

Step No.	Action	Process Name	Material	Depth	Mask
1	Wafer Setup	Wafer Setup1	Silicon / Si_XTAL	GeneralWafer...	
2	Sequence	FEOL	–	–	–
2.1 Fin Formation					
2.1	Sequence	Fin	–	–	–
2.1.1	Conformal Deposition	DEP PadOx	SiO_2	FinPadOxThick...	
2.1.2	Film Thickness	MEA THK PadOx	–	–	BB2
2.1.3	Conformal Deposition	DEP PadNit	Si_3N_4	FinPadNitThick...	
2.1.4	Film Thickness	MEA THK PadNit	–	–	BB2
2.1.5	Conformal Deposition	DEP Mandrel Si...	Silicon / Si_Am...	FinMandrelSiTh...	

Step No.	Action	Process Name	Material	Depth	Mask
2.1.6	Planarizing	DEP MandODL	Organics / ODL	LithoODLThick...	
2.1.7	Conformal Deposition	DEP MandSiARC	Silicon / SiARC	LithoARCThick...	
2.1.8	Planarizing	DEP MandRes	Organics / Resist	FinMandrelResi...	
2.1.9	Expose Material	LTH Mand	Organics / Resist	FinMandrelResi...	
2.1.10	Critical Dimension	MEA CD Mand...	–	–	CD_MAND
2.1.11	Sequence	RIE Mand	–	–	–
2.1.11.1	Legacy Basic Etch	RIE SiARC	Nitrides / Organics	28	–
2.1.11.2	Legacy Basic Etch	RIE ODL	Nitrides / Organics	EtchDepthODL	
2.1.11.3	Sequence	RIE PolyPick...	–	–	–
2.1.12	Critical Dimension	MEA CD MandFin	–	–	CD_MAND
2.1.13	Elevation Control	MEA SH MandFin	–	–	CD_MAND
2.1.14	Legacy Basic Etch	RIE MandRes	Nitrides / Organics	200	–
2.1.15	Conformal Deposition	DEP FinSpcrOx	SiO_2 LTO	FinSpcrOxThick...	
2.1.16	Sequence	RIE FinSpcrOx...	–	–	–
2.1.16.1	Legacy Basic Etch	RIE FinSpcrOx	Nitrides / Organics	FinSpcrOxThick...	
2.2 Fin Cut					
2.2	Sequence	FinCut	–	–	–
2.2.1	Planarizing	DEP FinCutRes	Organics / Resist	250	–
2.2.2	Expose Material	LTH FinCut	Organics / Resist	400	Active
2.2.3	Legacy Basic Etch	RIE FinCutSpcrOx	Nitrides / Organics	30	–
2.2.4	Remove Material	RIE FinCutRes	Organics / Resist	–	–
2.2.5	Legacy Basic Etch	RIE MandSiRem	Nitrides / Organics	70	–
2.2.6	Legacy Basic Etch	RIE PadNit	Nitrides / Organics	35	–

Step No.	Action	Process Name	Material	Depth	Mask
2.2.7	Legacy Basic Etch	RIE PlanActRes	Nitrides / Organics	200	–
2.2.8	Legacy Basic Etch	RIE PadOx	Nitrides / Organics	5	–
2.2.9	Sequence	RIE FinSi - Pick...	–	–	–
2.2.9.1	Sequence	RIE FinSi - Simple	–	–	–
2.2.10	Line to Line	MEA SP FinNN...	–	–	SP_FINNP
2.3 Well Formation					
2.3	Sequence	Wells	–	–	–
2.3.1	Conformal Deposition	DEP Screen Oxide	SiO_2	3.5	–
2.3.2	Conformal Deposition	DEP PSPD Poly	Silicon / Si_Am...	10	–
2.3.3	Conformal Deposition	DEP PWell Resist	Organics / Resist	400	–
2.3.4	Expose Material	LTH NWell	Organics / Resist	0	NWell
2.3.5	Legacy Basic Etch	RIE PSPD Poly	Metals / Nitrides	22	–
2.3.6	Remove Material	RIE Resist	Organics / Resist	–	–
2.3.7	Conformal Deposition	DEP NSPD Poly	Silicon / Si_Am...	18	–
2.3.8	Legacy Basic Etch	RIE NSPD Poly	Metals / Nitrides	80	–
2.3.9	Diffusion	RTP Well Drive	Nitrides / Oxides...	–	–
2.3.10	Legacy Basic Etch	RIE SPD Poly	Metals / Nitrides	24	–
2.4 STI Formation					
2.4	Sequence	STI	–	–	–
2.4.1	Conformal Deposition	DEP STI HDP	SiO_2	STIInitialDepTh...	
2.4.2	Polish	CMP HDP	–	–	–
2.4.3	Legacy Basic Etch	RIE STI Recess1	Nitrides / Organics	10	–
2.4.4	Wet Etch	WET HDP Clean	Nitrides / Organics	2	–
2.4.5	Legacy Basic Etch	WET PadNit Strip	Nitrides / Organics	FinPadNitThick...	

Step No.	Action	Process Name	Material	Depth	Mask
2.4.6	Legacy Basic Etch	RIE STI Recess2	–	STIfinRecess	–
2.5 Gate Formation					
2.5	Sequence	Gate	–	–	–
2.5.1	Conformal Deposition	DEP SacOx	SiO_2	GateSacOx	–
2.5.2	Film Thickness	MEA SacOx THK	–	–	CD_MAND
2.5.3	Conformal Deposition	DEP SacGate	Silicon / Si_Am...	GateSacGateHei...	
2.5.4	Polish	CMP SacGate	–	–	–
2.5.5	Straight Deposition	DEP GateCapNit	Si_3N_4	GateCapNitThic...	
2.5.6	Conformal Deposition	DEP GateCapOx	SiO_2 LTO	5	–
2.5.7	Planarizing	DEP GateODL	Organics / ODL	LithoODLThick...	
2.5.8	Conformal Deposition	DEP GateSiARC	Silicon / SiARC	LithoARCThick...	
2.5.9	Planarizing	DEP GateRes	Organics / Resist	GateResist	–
2.5.10	Expose Material	LTH Gate	Organics / Resist	GateResist+100	GateUncut
2.5.11	Sequence	RIE Gate Hard...	–	–	–
2.5.11.1	Legacy Basic Etch	RIE SiARC	Nitrides / Organics	LithoARCThick...	
2.5.11.2	Legacy Basic Etch	RIE ODL	Nitrides / Organics	EtchDepthODL	
2.5.11.3	Legacy Basic Etch	RIE Oxide BT	–	8	–
2.5.11.4	Sequence	RIE Nitride - Pick...	–	–	–
2.5.11.4.1	Legacy Basic Etch	RIE Nitride	Nitrides / Nit...	GateCapNitThic...	
2.5.12	Legacy Basic Etch	RIE GateRes	Nitrides / Organics	200	–
2.5.13	Planarizing	DEP GateCutODL	Organics / ODL	LithoODLThick...	
2.5.14	Conformal Deposition	DEP GateCutSiARC	Silicon / SiARC	LithoARCThick...	
2.5.15	Planarizing	DEP GateCutRes	Organics / Resist	GateResist	–

Step No.	Action	Process Name	Material	Depth	Mask
2.5.16	Expose Material	LTH GateCut	Organics / Resist	150	GateCut
2.5.17	Sequence	RIE GateCut	–	–	–
2.5.17.1	Legacy Basic Etch	RIE GateCut SiA...	Nitrides / Organics	LithoARCThick...	–
2.5.17.2	Legacy Basic Etch	RIE GateCut ODL	Nitrides / Organics	EtchDepthODL	–
2.5.17.3	Legacy Basic Etch	RIE GateCut Ox...	–	8	–
2.5.17.4	Legacy Basic Etch	RIE GateCut Nit...	Nitrides / Nit...	GateCapNitThic...	–
2.5.18	Legacy Basic Etch	RIE GateCutRes	Nitrides / Organics	200	–
2.5.19	Sequence	RIE GateFinal - ...	–	–	–
2.5.20	Wet Clean	WET Clean	Organics / Oxides...	1	–
2.6 Source/Drain Formation					
2.6	Sequence	Source/Drain	–	–	–
2.6.1	Sequence	PSD	–	–	–
2.6.1.1	Conformal Deposition	DEP PSpacerOx	SiO_2 LTO	SourceDrainOx...	–
2.6.1.2	Conformal Deposition	DEP PSpacerNit	Si_3N_4	SourceDrainNit...	–
2.6.1.3	Planarizing	DEP PSpacerRes	Organics / Resist	180	–
2.6.1.4	Expose Material	LTH PSpacer	Organics / Resist	600	NWell
2.6.1.5	Legacy Basic Etch	RIE PSpacer	Nitrides / Nit...	SourceDrainNit...	–
2.6.1.6	Sequence	RIE PSD Recess...	–	–	–
2.6.1.7	Legacy Basic Etch	RIE PSpacerRes	Nitrides / Organics	300	–
2.6.1.8	Crystal Etch	WET PRecess	Oxides / Silicon	10	–
2.6.1.9	Selective Epitaxy	EPI SiGe	Silicon / SiGe	SourceDrainSiG...	–
2.6.2	Sequence	NSD	–	–	–
2.6.2.1	Conformal Deposition	DEP NSpacerNit	Si_3N_4	SourceDrainExtr...	–
2.6.2.2	Planarizing	DEP NSpacerRes	Organics / Resist	180	–

Step No.	Action	Process Name	Material	Depth	Mask
2.6.2.3	Expose Material	LTH NSpacer	Organics / Resist	600	NWell
2.6.2.4	Legacy Basic Etch	RIE NSpacer	Nitrides / Nit...	SourceDrainNit...	
2.6.2.5	Sequence	RIE NSD Recess...	–	–	–
2.6.2.6	Legacy Basic Etch	RIE NSpacer-Res	Nitrides / Organics	300	–
2.6.2.7	Crystal Etch	WET NRecess	Oxides / Silicon	10	–
2.6.2.8	Selective Epitaxy	EPI SiC	Silicon / SiC	SourceDrainSiC...	
2.6.3	Diffusion	RTA	Silicon / SiC / Si...	–	–

(a) Spacer Deposition and Patterning Process Steps:

The following table shows the important process steps involved in spacer deposition and patterning in the 14 nm FinFET structure.

Sr. No.	Process Step	Action Performed
1	2.1.15	Conformal deposition of silicon dioxide (SiO_2) to form the fin spacer oxide layer.
2	2.1.16	Etching of the deposited fin spacer oxide to retain the required spacer structure.
3	2.6.1.1	Deposition of silicon dioxide (SiO_2) for P-type spacer formation.
4	2.6.1.2	Deposition of silicon nitride (Si_3N_4) over the P-type spacer structure.
5	2.6.1.5	Etching of the deposited P-type spacer nitride layer.
6	2.6.2.1	Deposition of silicon nitride (Si_3N_4) for N-type spacer formation.
7	2.6.2.4	Etching of the deposited N-type spacer nitride layer.

Table 3: Spacer deposition and patterning process steps in 14 nm FinFET

Spacer Deposition and Patterning Images:

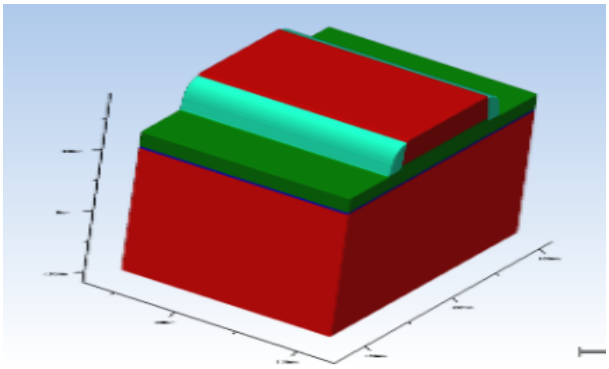


Figure 6: Fin Spacer Oxide Deposition and Etching (Steps 1 and 2)

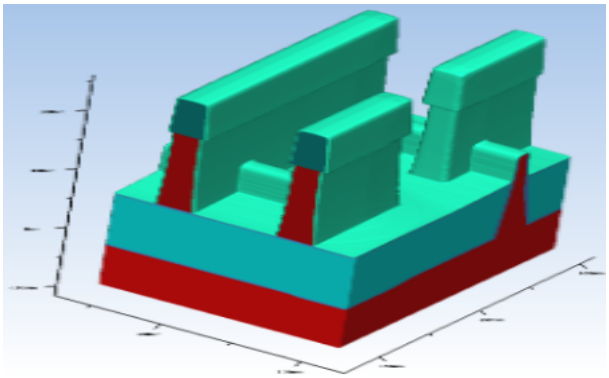


Figure 7: P-Type Spacer Nitride Deposition (Step 4)

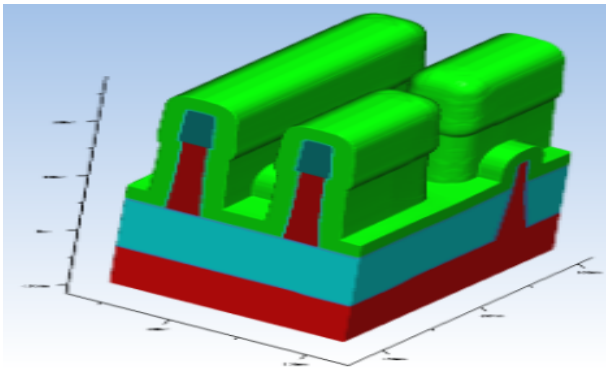


Figure 8: P-Type Spacer Nitride Etching (Step 5)

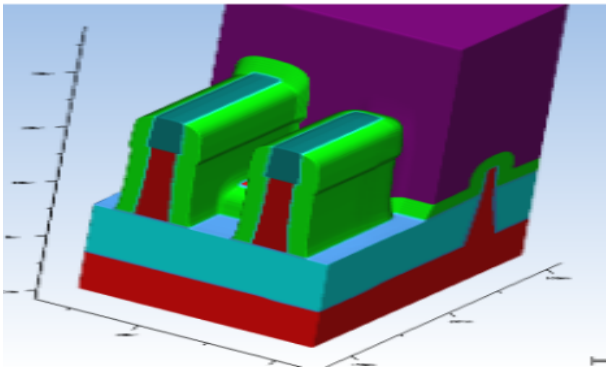


Figure 9: N-Type Spacer Nitride Deposition (Step 6)

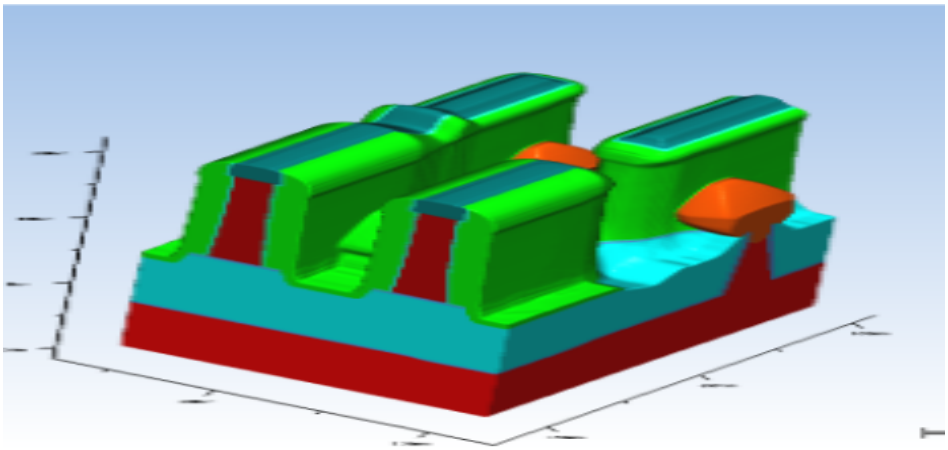


Figure 10: N-Type Spacer Etching (Step 7)

(b) Materials Used for Spacer:

In the 14 nm FinFET structure, two dielectric materials are used for spacer formation:

- **Silicon Dioxide (SiO_2):** Used for fin spacer oxide formation and P-type spacer formation.
- **Silicon Nitride (Si_3N_4):** Used for P-type and N-type spacer formation.

5. OBSERVATIONS

1. Spacer formation was observed on the vertical sidewalls of the gate structure after conformal deposition and anisotropic etching.
2. In Question 1, the simulated CD width was approximately 7.0117 nm. The CD gradually decreased along the spacer height, and the calculated CD loss at 50% of the spacer height was approximately 0.3626 nm.
3. In Question 2, the spacer deposition and patterning steps in the 14 nm FinFET process flow were identified.
4. Silicon dioxide (SiO_2) and silicon nitride (Si_3N_4) were identified as the dielectric materials used for spacer formation.
5. Separate spacer formation steps were observed for fin, P-type, and N-type regions.

6. CONCLUSION

The spacer formation process was studied using the Spacer.zam project and the inbuilt 14nmFinFET.zam example in Semulator3D. The importance of conformal deposition and anisotropic etching in forming self-aligned spacers was understood.

The CD variation along the spacer height was analyzed, and the CD loss at 50% of the spacer height was calculated. The deposition and patterning steps in the 14 nm FinFET process were also identified. The use of silicon dioxide and silicon nitride as spacer materials was studied. Overall, this assignment provided an understanding of spacer formation and its importance in MOSFET and FinFET fabrication.

Signature: